Lesson 31: Robbins–Monro and the ODE Method

AI/ML Mathematics

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1 Motivation and Overview

Reinforcement learning algorithms optimize under uncertainty: an agent seeks an optimal policy when the environment dynamics are unknown and the only information arrives through noisy, sequential interactions. The mathematical framework that captures this setting is *stochastic approximation* (SA) — finding the root (or optimum) of a function that can only be observed through noisy samples.

The field was born with the seminal 1951 paper of Robbins and Monro [1951], who proposed an iterative scheme for finding the root of a regression function from noisy observations. The theoretical foundations were deepened by Kushner and Clark, Ljung, and Borkar [2008], who established the powerful *ODE method*: under appropriate conditions, a stochastic approximation algorithm converges to the same limit as the solution of an associated ordinary differential equation.

SA as the Engine of RL

Every major RL update rule is a stochastic approximation iteration:

- **TD(0):** $V(s_n) \leftarrow V(s_n) + \alpha_n[R_{n+1} + \gamma V(s_{n+1}) - V(s_n)]$
- **Q-learning:** $Q(s_n, a_n) \leftarrow Q(s_n, a_n) + \alpha_n[R_{n+1} + \gamma \max_{a'} Q(s_{n+1}, a') - Q(s_n, a_n)]$
- **REINFORCE:** $\theta_{n+1} = \theta_n + \alpha_n \widehat{\nabla J(\theta_n)}$
- **Actor-Critic:** critic and actor update at *different rates* — two-timescale SA (Lesson 32).

The convergence theorems in this lesson provide the theoretical foundation for understanding *when* and *why* these algorithms converge.

Prerequisites. Probability spaces, conditional expectation, martingale difference sequences (Lesson 8), the law of large numbers, and basic ODE theory (existence, uniqueness, Lyapunov stability).

2 References

The core references for this lesson are:

- Borkar [2008], Chapters 2–3: the definitive modern treatment of SA via the ODE method, including the Borkar–Meyn stability theorem.
- Kushner and Yin [2003], Chapters 1–5: the classical Kushner–Clark approach to SA convergence, with extensive applications.
- Robbins and Monro [1951]: the original paper introducing the Robbins–Monro iteration.
- Borkar and Meyn [2000]: the key paper establishing the Borkar–Meyn stability criterion and its application to RL.
- Benveniste et al. [2012], Chapters 1–2: a rigorous treatment of SA using the ODE method, with applications to adaptive algorithms.

3 The Robbins–Monro Algorithm

3.1 Problem Formulation

Suppose we wish to find a root of a function $h : \mathbb{R}^d \to \mathbb{R}^d$, i.e., a point $x^* \in \mathbb{R}^d$ such that $h(x^*) = 0$. The difficulty is that we cannot evaluate h directly. Instead, for each query point x,

we observe a noisy sample

$$H(x, Y) \quad \text{where} \quad h(x) = \mathbb{E}[H(x, Y)]$$

and Y is a random element representing the noise source.

Definition 3.1 (Step-Size Sequence (Robbins–Monro Conditions))**.** A *step-size sequence* (or *learning rate schedule*) is a deterministic sequence $\{a_n\}_{n\geq 0}$ of positive real numbers satisfying the *Robbins–Monro conditions*:

(i) $\displaystyle\sum_{n=0}^{\infty} a_n = \infty$ (steps are large enough to reach any target),

(ii) $\displaystyle\sum_{n=0}^{\infty} a_n^2 < \infty$ (steps are small enough that noise averages out).

Remark 3.2. Condition (i) ensures the iterates can travel an unbounded total distance. Condition (ii) controls accumulated noise variance (proportional to $\sum a_n^2 \operatorname{Var}(\text{noise})$). The canonical choice is $a_n = c/n$; more generally, $a_n = c/n^\alpha$ works for $\alpha \in (1/2, 1]$.

3.2 The Iteration

Definition 3.3 (Robbins–Monro Iteration)**.** Given an initial point $x_0 \in \mathbb{R}^d$, the *Robbins–Monro iteration* is

$$x_{n+1} = x_n + a_n H(x_n, Y_n), \quad n \geq 0, \tag{1}$$

where $\{a_n\}$ satisfies Definition 3.1 and $Y_0, Y_1, \ldots$ are noise samples.

We decompose the update into "signal" and "noise." Define the *martingale difference noise*

$$M_{n+1} \coloneqq H(x_n, Y_n) - h(x_n). \tag{2}$$

Then the iteration becomes

$$x_{n+1} = x_n + a_n\big[h(x_n) + M_{n+1}\big]. \tag{3}$$

This is the canonical form we use throughout: the iterate moves in the direction $h(x_n)$ (the "drift") plus additive noise M_{n+1}.

3.3 Standing Assumptions

Definition 3.4 (Standing Assumptions for Robbins–Monro)**.** Let $\mathcal{F}_n = \sigma(x_0, Y_0, \ldots, Y_{n-1})$ be the natural filtration. We assume:

(A1) **Step sizes:** $\{a_n\}$ satisfies Definition 3.1.

(A2) **Martingale difference noise:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = 0$ a.s. for all n.

(A3) **Bounded noise variance:** There exists $K > 0$ such that $\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \leq K(1+\|x_n\|^2)$ a.s.

(A4) **Stability (monotonicity):** There exists a unique root $x^* \in \mathbb{R}^d$ with $h(x^*) = 0$, and $\langle x - x^*, h(x)\rangle < 0$ for all $x \neq x^*$.

(A5) **Linear growth:** $\|h(x)\| \leq K(1 + \|x\|)$ for all x.

Remark 3.5. Assumption (A4) requires that h points "inward" toward x^*. Geometrically, the angle between $x - x^*$ and $h(x)$ is always obtuse. This is satisfied when $h = -\nabla f$ for a strictly convex f with minimum at x^*. For non-convex problems (e.g., policy gradient in RL), this condition fails, motivating the ODE method of Section 4.

3.4 Robbins–Siegmund Theorem

The convergence proof relies on a classical supermartingale result.

Theorem 3.6 (Robbins–Siegmund). *Let $\{V_n\}$, $\{\alpha_n\}$, $\{\beta_n\}$, $\{u_n\}$ be nonnegative $\mathcal{F}_n$-adapted sequences satisfying*

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1+\alpha_n)V_n - \beta_n + u_n$$

with $\sum \alpha_n < \infty$ and $\sum u_n < \infty$ a.s. Then:

(i) V_n converges a.s. to a finite limit.

(ii) $\sum \beta_n < \infty$ a.s.

Proof. Define $W_n = V_n \prod_{k=0}^{n-1}(1+\alpha_k)^{-1}$. Since $\sum \alpha_k < \infty$, the product $P_n = \prod_{k=0}^{n-1}(1+\alpha_k)$ converges to a finite limit, so $1/P_n$ is bounded away from zero. Dividing the hypothesis by P_{n+1} and rearranging shows that $Z_n = W_n + \sum_{k=n}^{\infty} u_k/P_k$ is a nonnegative supermartingale. By the supermartingale convergence theorem, Z_n converges a.s., hence W_n and V_n converge a.s. Summing the inequality yields $\sum \beta_n/P_n < \infty$, and since $1/P_n$ is bounded below, $\sum \beta_n < \infty$ a.s. □

3.5 Convergence Theorem

Theorem 3.7 (Robbins–Monro Convergence). *Under assumptions (A1)–(A5), the Robbins–Monro iteration (1) satisfies $x_n \xrightarrow{a.s.} x^*$.*

Proof. Define $e_n = x_n - x^*$. Expanding $\|e_{n+1}\|^2$ and taking conditional expectation (using (A2)):

$$\mathbb{E}[\|e_{n+1}\|^2 \mid \mathcal{F}_n] = \|e_n\|^2 + 2a_n\langle e_n, h(x_n)\rangle + a_n^2\mathbb{E}[\|h(x_n) + M_{n+1}\|^2 \mid \mathcal{F}_n].$$

Using $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, (A3), (A5), and $\|x_n\|^2 \leq 2\|e_n\|^2 + 2\|x^*\|^2$, we get

$$\mathbb{E}[\|e_{n+1}\|^2 \mid \mathcal{F}_n] \leq (1 + Ca_n^2)\|e_n\|^2 + 2a_n\langle e_n, h(x_n)\rangle + Ca_n^2$$

for some $C > 0$. Set $V_n = \|e_n\|^2$, $\alpha_n = Ca_n^2$, $\beta_n = -2a_n\langle e_n, h(x_n)\rangle$, $u_n = Ca_n^2$. By (A4), $\beta_n > 0$ when $x_n \neq x^*$. Since $\sum a_n^2 < \infty$, the Robbins–Siegmund theorem gives: (i) V_n converges a.s. to some $L \geq 0$, and (ii) $\sum \beta_n < \infty$ a.s.

If $L > 0$ with positive probability, then $\|e_n\| \to \sqrt{L} > 0$, so by (A4) there exists $\delta > 0$ with $\beta_n \geq 2\delta a_n$ eventually, giving $\sum \beta_n \geq 2\delta \sum a_n = \infty$, contradicting (ii). Hence $L = 0$ a.s. □

Robbins–Monro Convergence

Under a monotonicity condition on h (the drift points toward the root x^*), martingale difference noise with bounded conditional variance, and the step-size conditions $\sum a_n = \infty$, $\sum a_n^2 < \infty$, the iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ converges almost surely to x^*.

SGD as Robbins–Monro

Stochastic gradient descent for minimizing $f(\theta) = \mathbb{E}_Z[\ell(\theta, Z)]$ is the Robbins–Monro iteration with $h(\theta) = -\nabla f(\theta)$ and $H(\theta, Z) = -\nabla_\theta \ell(\theta, Z)$. Under strict convexity of f, the monotonicity condition (A4) holds for $-\nabla f$, so Theorem 3.7 guarantees $\theta_n \xrightarrow{\text{a.s.}} \theta^*$.

4 The ODE Method

The Robbins–Monro analysis requires the strong monotonicity condition (A4), which fails for many RL algorithms (e.g., policy gradient on non-concave objectives). The *ODE method*, developed by Ljung (1977), Kushner and Clark (1978), and systematized by Borkar [2008], replaces monotonicity with a more natural condition: convergence of the SA iteration follows from the stability of an associated ordinary differential equation.

4.1 The Associated ODE

Definition 4.1 (Associated ODE)**.** Given the SA iteration (3), the *associated ODE* is

$$\dot{x}(t) = h(x(t)), \quad x(0) = x_0. \tag{4}$$

The key insight is that for small step sizes, the SA iteration $x_{n+1} \approx x_n + a_n h(x_n)$ behaves like an Euler discretization of the ODE (4) with time step a_n, plus noise that averages out.

4.2 Interpolated Trajectory

Definition 4.2 (Interpolated Trajectory)**.** Define the time instants $t_0 = 0$ and $t_n = \sum_{k=0}^{n-1} a_k$ for $n \geq 1$. Note that $t_n \to \infty$ by condition (i) of Definition 3.1. The *interpolated trajectory* $\bar{x} : [0, \infty) \to \mathbb{R}^d$ is the piecewise constant function

$$\bar{x}(t) = x_n \quad \text{for } t \in [t_n, t_{n+1}).$$

4.3 ODE Method Assumptions

Definition 4.3 (ODE Method Assumptions)**.** We modify the standing assumptions as follows:

(B1) **Step sizes:** As in (A1).

(B2) **Lipschitz drift:** $h : \mathbb{R}^d \to \mathbb{R}^d$ is Lipschitz continuous: $\|h(x) - h(y)\| \leq L\|x - y\|$ for all x, y and some $L > 0$.

(B3) **Martingale difference noise:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = 0$ a.s. and $\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \leq K(1 + \|x_n\|^2)$ a.s.

(B4) **Stability (boundedness):** $\sup_n \|x_n\| < \infty$ a.s.

(B5) **Asymptotic stability of the ODE:** The ODE (4) has a globally asymptotically stable equilibrium x^* (i.e., $h(x^*) = 0$ and $x(t) \to x^*$ as $t \to \infty$ for every initial condition).

Remark 4.4 (The Boundedness Assumption (B4)). Assumption (B4) is the most delicate. In practice, one either (a) proves it from properties of h via the Borkar–Meyn theorem (Section 5), (b) uses projection to a compact set, or (c) verifies it empirically. For many RL algorithms, boundedness follows from the structure of the MDP (e.g., bounded rewards and $\gamma < 1$).

4.4 The Noise Vanishes Asymptotically

Lemma 4.5 (Noise Term Vanishes)**.** *Under (B1), (B3), and (B4), the martingale $Z_n = \sum_{k=0}^{n-1} a_k M_{k+1}$ converges a.s. In particular, for any $T > 0$,*

$$\sup_{t_n \leq s \leq t_n + T} \left\| \sum_{k=n}^{m(s)-1} a_k M_{k+1} \right\| \to 0 \quad \textit{a.s. as } n \to \infty,$$

where $m(s) = \min\{j : t_j > s\}$.

Proof. Z_n is a martingale with $\mathbb{E}[\|Z_n\|^2] = \sum_{k=0}^{n-1} a_k^2 \mathbb{E}[\|M_{k+1}\|^2] \le K(1+C)\sum a_k^2 < \infty$ (using (B4) to bound $\mathbb{E}[\|x_k\|^2] \le C$). By the martingale convergence theorem, Z_n converges a.s. The tail sum $Z_{m(s)} - Z_n \to 0$ uniformly over $s \in [t_n, t_n + T]$ since convergent series have vanishing tails. □

4.5 Convergence via the ODE Method

Theorem 4.6 (Convergence via the ODE Method)**.** *Under assumptions (B1)–(B5), the SA iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ satisfies $x_n \xrightarrow{a.s.} x^*$.*

Proof. The proof proceeds in three steps.

Step 1: Shifted interpolated trajectories. For each $n \ge 0$, define the shifted interpolated trajectory $x^{(n)} : [0, \infty) \to \mathbb{R}^d$ by

$$x^{(n)}(0) = x_n, \quad x^{(n)}(t) = x_{n+k} \text{ for } t \in [t_{n+k} - t_n,\, t_{n+k+1} - t_n), \quad k \ge 0.$$

Step 2: Convergence to ODE solutions. By (B4) and (B2), both $\|x_n\|$ and $\|h(x_n)\|$ are bounded a.s. The trajectory increments satisfy $\|x^{(n)}(t) - x^{(n)}(s)\| \le C|t-s| + \varepsilon_n$ where $\varepsilon_n \to 0$ by Lemma 4.5, giving equicontinuity on $[0, T]$. By the Arzela–Ascoli theorem, every subsequence has a uniformly convergent further subsequence with limit $x^{(\infty)}$. The SA iteration gives

$$x^{(n)}(t) = x^{(n)}(0) + \int_0^t h(x^{(n)}(s))\, ds + \varepsilon_n(t),$$

where $\varepsilon_n(t) \to 0$ (discretization error vanishes since $a_n \to 0$; noise vanishes by Lemma 4.5). Passing to the limit via Lipschitz continuity of h shows $x^{(\infty)}$ solves the ODE.

Step 3: Convergence to the equilibrium. By (B5), every ODE solution converges to x^*. The set of limit points of $\{x_n\}$ is nonempty (by (B4)), compact, connected, and ODE-flow-invariant (by Step 2). The only such set attracted to x^* under global asymptotic stability is $\{x^*\}$ itself: if $\bar{x} \ne x^*$ were a limit point, the ODE trajectory from $\bar{x}$ would enter any neighborhood of x^*, and Lyapunov stability would prevent escape, making x^* the unique limit point. □

The ODE Method

A stochastic approximation iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ with decreasing step sizes converges to the set of asymptotically stable equilibria of the ODE $\dot{x} = h(x)$, provided the iterates remain bounded and the noise is a martingale difference sequence. This is the primary tool for analyzing RL algorithm convergence.

TD(0) via the ODE Method

For TD(0) in tabular form, the mean field is $h(V) = r^\pi + \gamma P^\pi V - V = (\gamma P^\pi - I)V + r^\pi$. The associated ODE $\dot{V} = (\gamma P^\pi - I)V + r^\pi$ is a linear system. The matrix $A = \gamma P^\pi - I$ has all eigenvalues with real part at most $\gamma - 1 < 0$ (since P^π has spectral radius 1), so A is Hurwitz. The unique equilibrium is $V^\pi = (I - \gamma P^\pi)^{-1} r^\pi$, the true value function, and it is globally asymptotically stable. By Theorem 4.6, TD(0) converges a.s. to V^π.

5 The Borkar–Meyn Stability Theorem

Assumption (B4) (boundedness of iterates) is often the hardest to verify. The Borkar–Meyn theorem [Borkar and Meyn, 2000] provides a systematic way to establish it by examining the behavior of h at infinity.

Definition 5.1 (Borkar–Meyn Assumptions)**.** In addition to (B1)–(B3), assume:

(C1) h is Lipschitz and the *scaled limit* $h_\infty(x) := \lim_{r\to\infty} h(rx)/r$ exists uniformly on compact sets.

(C2) The ODE $\dot{x} = h_\infty(x)$ has the origin as its unique globally asymptotically stable equilibrium.

Remark 5.2. For an affine function $h(x) = Ax + b$, the scaled limit is $h_\infty(x) = \lim_{r\to\infty}(A(rx) + b)/r = Ax$. So (C2) reduces to requiring that A is Hurwitz. For nonlinear functions that behave linearly at infinity, h_∞ captures the dominant asymptotic behavior.

Theorem 5.3 (Borkar–Meyn Stability)**.** *Under (B1)–(B3) and (C1)–(C2), the iterates* $\{x_n\}$ *are bounded almost surely:* $\sup_n \|x_n\| < \infty$ *a.s.*

Proof. Suppose for contradiction $\|x_n\| \to \infty$ along a subsequence with positive probability. Let $r_n = \max(\|x_0\|, \ldots, \|x_n\|)$ and $\hat{x}_n = x_n/r_n$. Dividing the SA iteration by r_{n+1} and using (C1) ($h(r_n\hat{x}_n)/r_n \to h_\infty(\hat{x}_n)$), the rescaled trajectory converges along subsequences to solutions of $\dot{x} = h_\infty(x)$ (by the same interpolation argument as in Theorem 4.6). But $\|\hat{x}_n\| \le 1$ and $\|\hat{x}_n\| = 1$ infinitely often, so the limiting trajectory is confined to the unit ball and visits its boundary. By (C2), every solution of $\dot{x} = h_\infty(x)$ converges to the origin, giving a contradiction. □

Borkar–Meyn Stability Criterion

To verify iterate boundedness for SA, examine the "behavior of h at infinity": compute $h_\infty(x) = \lim_{r\to\infty} h(rx)/r$ and check that $\dot{x} = h_\infty(x)$ has a stable origin. For linear SA ($h(x) = Ax + b$), this reduces to checking that A is Hurwitz.

Step-Size Schedules in Practice

Common RL step-size strategies and their SA interpretation:

- **Polynomial decay:** $a_n = c/(n+d)^\alpha$, $\alpha \in (1/2, 1]$. Satisfies both conditions.
- **Constant step size:** $a_n = \alpha$. Violates $\sum a_n^2 < \infty$; iterates fluctuate in an $O(\alpha)$ neighborhood of x^*. Preferred for non-stationary environments.
- **Warmup + decay:** Used in PPO, SAC. No exact convergence guarantee but good empirical performance.

The Borkar–Meyn theorem explains why constant-step-size RL remains bounded: if h_∞ is stable, iterates cannot escape to infinity.

6 Stochastic Root-Finding

Example 6.1 (Mean Estimation as SA)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. The iteration $\theta_{n+1} = \theta_n + a_n(X_{n+1} - \theta_n)$ has $h(\theta) = \mu - \theta$, $x^* = \mu$, and noise $M_{n+1} = X_{n+1} - \mu$. Monotonicity (A4) holds: $(\theta - \mu)(\mu - \theta) = -(\theta - \mu)^2 < 0$. The associated ODE $\dot{\theta} = \mu - \theta$ has solution $\theta(t) = \mu + (\theta_0 - \mu)e^{-t}$ and Jacobian $A = -1$ (Hurwitz).

Example 6.2 (Quantile Estimation)**.** Let $X_1, X_2, \ldots$ be i.i.d. with continuous strictly increasing CDF F. To find the p-quantile $q_p = F^{-1}(p)$:

$$\theta_{n+1} = \theta_n + a_n(p - \mathbf{1}(X_{n+1} \le \theta_n)).$$

Here $h(\theta) = p - F(\theta)$, $x^* = q_p$, and the noise $M_{n+1} = F(\theta_n) - \mathbf{1}(X_{n+1} \le \theta_n)$ has conditional mean zero. The monotonicity condition (A4) holds because F is increasing: if $\theta > q_p$ then $F(\theta) > p$ so $(\theta - q_p)(p - F(\theta)) < 0$, and vice versa.

Example 6.3 (Q-learning Update as SA)**.** In Q-learning, $h(Q)(s,a) = r(s,a)+\gamma \sum_{s'} P(s'|s,a) \max_{a'} Q(s',a') - Q(s,a)$. This is *nonlinear* SA. The associated ODE $\dot{Q} = \mathcal{T}^* Q - Q$ involves the Bellman optimality operator, a γ-contraction in the sup-norm. The Lyapunov function $V(Q) = \|Q - Q^*\|_\infty$ satisfies $\dot{V} \leq -(1-\gamma)V$, so Q^* is globally asymptotically stable. By the ODE method, Q-learning converges a.s. to Q^*.

7 Exercises

Exercise 7.1 (Step-Size Conditions (*))**.** For each of the following, determine whether both Robbins–Monro conditions hold:

(a) $a_n = 1/n$

(b) $a_n = 1/n^2$

(c) $a_n = 1/\sqrt{n}$

(d) $a_n = c/(n+d)$ for constants $c, d > 0$

(e) $a_n = 1/(n \log n)$ for $n \geq 2$

(f) $a_n = 1/n^{3/4}$

For each case that fails, explain which condition fails and the practical consequence for an SA algorithm.

Exercise 7.2 (Scalar Robbins–Monro (**))**.** Consider $x_{n+1} = x_n + a_n(c - x_n + \xi_{n+1})$ where $c \in \mathbb{R}$ and $\{\xi_n\}$ are i.i.d. with $\mathbb{E}[\xi_n] = 0$, $\mathbb{E}[\xi_n^2] = \sigma^2$.

(a) Identify $h(x)$ and x^*.

(b) Verify the monotonicity condition (A4).

(c) Compute the Jacobian $A = h'(x^*)$ and verify A is Hurwitz.

(d) Write the associated ODE and solve it explicitly.

(e) Show that $\|x_n - x^*\|^2$ satisfies a recursion compatible with the Robbins–Siegmund theorem.

Exercise 7.3 (ODE for Q-Learning (**))**.** Consider a two-state, two-action MDP with $\mathcal{S} = \{1,2\}$, $\mathcal{A} = \{L, R\}$, $\gamma = 0.9$, deterministic transitions $P(2|1,L) = 1$, $P(1|1,R) = 1$, $P(1|2,L) = 1$, $P(2|2,R) = 1$, and rewards $r(1,L) = 1$, $r(1,R) = 0$, $r(2,L) = 0$, $r(2,R) = 2$.

(a) Compute $Q^*(s,a)$ for all (s,a) by solving the Bellman optimality equation.

(b) Write the associated ODE $\dot{Q} = h(Q)$ as a 4-dimensional system.

(c) Verify that Q^* is the unique equilibrium.

Exercise 7.4 (Lyapunov Function for Contraction (**))**.** Let $T : \mathbb{R}^d \to \mathbb{R}^d$ be a γ-contraction in the sup-norm with fixed point x^*. Consider the ODE $\dot{x} = Tx - x$.

(a) Show that $V(x) = \|x - x^*\|_\infty$ satisfies $\dot{V} \leq -(1-\gamma)V$.

(b) Conclude that x^* is globally asymptotically stable.

(c) Apply this to $T = \mathcal{T}^*$ (Bellman optimality operator) to verify the ODE analysis for Q-learning.

Exercise 7.5 (Borkar–Meyn for Linear SA (**))**.** For linear SA with $h(x) = Ax + b$:

(a) Compute $h_\infty(x)$ and state the Borkar–Meyn condition (C2) in terms of the eigenvalues of A.

(b) Verify that the condition is equivalent to A being Hurwitz.

(c) Let $A = \begin{pmatrix} -2 & 1 \\ 0 & -3 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Find the equilibrium $x^* = -A^{-1}b$ and verify boundedness.

Exercise 7.6 (SA with Markovian Noise (***))**.** In RL, noise is Markovian (from a Markov chain), not i.i.d. Suppose $\{Y_n\}$ is ergodic with stationary distribution μ, $h(x) = \mathbb{E}_\mu[H(x, Y)]$, and the Poisson equation $g(x, y) - Pg(x, y) = H(x, y) - h(x)$ has a solution g. Show how to decompose $H(x_n, Y_n) - h(x_n)$ into a martingale difference plus a telescoping term using g, recovering the ODE method assumptions.

8 Summary

- **Robbins–Monro conditions:** $\sum a_n = \infty$ and $\sum a_n^2 < \infty$. Canonical choice: $a_n = c/n$.
- **Robbins–Monro convergence:** Under monotonicity of h, bounded noise variance, and RM step sizes, $x_n \xrightarrow{\text{a.s.}} x^*$.
- **ODE method:** SA converges to stable equilibria of $\dot{x} = h(x)$, given bounded iterates and Lipschitz h. The primary tool for RL convergence proofs.
- **Borkar–Meyn:** Boundedness holds if $h_\infty(x) = \lim_{r\to\infty} h(rx)/r$ has a stable origin. For linear SA, reduces to A Hurwitz.
- **RL:** TD(0) is linear SA with Hurwitz $A = \gamma P^\pi - I$. Q-learning is nonlinear SA with contraction-based stability.

9 Further Reading

- Lesson 32: convergence rates, Polyak–Ruppert averaging, two-timescale SA, linear SA for TD(0) with function approximation.
- Borkar [2008]: complete theory including lock-in, differential inclusions, constrained SA.
- Kushner and Yin [2003]: weak convergence approach with applications.
- Benveniste et al. [2012]: SA with Markovian noise.

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